Methods in Java

Static, Non-Static, Parameterized and Return Type

Static Method :-.

- Belongs to the **class**, not the object.
- Can be called **without creating an object**.
- Often used for utility/helper methods.
- Can access only **static variables** directly.

- **Syntax :-**

```
static returnType methodName()
    {
        // block of code
    }
```

- **Example :-**

```
class Demo {
    public static void main(String[] args)
    {
        Demo.display();
    }

    static void display()
    {
        System.out.println("This is a Static Method");
    }
}
```

Non-Static Method :-

- Belongs to the **object** of the class.
- Must create an **object** to call it.
- Can access both **static** and **non-static** variables.

- **Syntax :-**

```
returnType methodName()
        {
                // block of code
        }
```

- **Example :-**

```
class Demo
{
    void show()
    {
        System.out.println("This is a Non-Static Method");
    }

    public static void main(String[] args)
    {
        Demo obj = new Demo();
        obj.show();
    }
}
```

Parameterized Method :-

- Takes inputs (parameters) **while calling.**
- Used when we need to **pass values for processing.**
- Parameters can be of **any data type.**

- **Syntax :-**

```
returnType methodName(dataType param1, dataType param2)
        {
                // block of code
        }
```

- **Example :-**

```
class Demo
{
    void greet(String name)
    {
        System.out.println("Hello, " + name);
    }

    public static void main(String[] args)
    {
        Demo obj = new Demo();
        obj.greet("Nikhil");
    }
}
```

Return Type Method :-

- Returns a value **after execution.**
- **return keyword** is used to send the **value back.**
- Return type can be **int**, **double**, **String**, etc.
- If **nothing is returned** → use **void.**

- **Syntax :-**

```
returnType methodName(parameters)
        {
                // block of code
                return value;
        }
```

- **Example :-**

```
class Demo
{
   int add(int a, int b)
   {
      return a + b;
   }

   public static void main(String[] args)
   {
      Demo obj = new Demo();
      int result = obj.add(5, 10);
      System.out.println("Sum = " + result);
   }
}
```

Implementation

Static Method :-

Input :-

```
1  package basicconcepts;
2
3  public class method_static {
4      public static void main(String[] args) {
5
6          display();
7  }
8
9  static void display()
10 {
11     System.out.println("Static Method");
12 }
13 }
```

Output :-

```
Static Method
```

Non-Static Method :-

Input :-

```java
package basicconcepts;

public class method_nonstatic {

    public static void main(String[] args) {

        //1.Creating Object
        method_nonstatic obj=new method_nonstatic();

        //2.Call method_nonststic method using object
        obj.add();
    }

    void add()
    {
        System.out.println("Addition is:" +(4+8));
    }
}
```

Output :-

```
Addition is:12
```

Parameterized Method :-

Input :-

```java
package basicconcepts;

public class method_parametrized {

    public static void main(String[] args) {

        substraction(40,20);

    }

    static void substraction(int n1, int n2)
    {
        System.out.println("Substraction is : "+(n1-n2));
    }
}
```

Output :-

```
Substraction is : 20
```